

MULS-Net: A Multilevel Supervised Network for Ship Tracking From Low-Resolution Remote-Sensing Image Sequences

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Abstract—Ship detection and tracking from remote-sensing image sequences has become an increasingly important research point. However, there are still many challenges for ship tracking from low-resolution remote-sensing image sequences: 1) the dim and small objects contain only a few shape and texture features, making it difficult to detect and track ships; 2) broken clouds often resemble ships, resulting in false tracking; and 3) the ship may be occluded by clouds leading to missed tracking. To address these challenges, we proposed a novel multilevel supervision network for ship tracking from low-resolution remote-sensing image sequences. First, we designed a gradient difference-guided object clarification network component to significantly improve the object saliency, which is also implemented based on the multi-frame correlation enhancement images to improve the feature strength of small targets in the input data. Second, to reduce the difficulty of completing complex tasks, a multilevel supervised network (MULS-Net) framework with multiple components was presented to improve the target clarity, detecting targets, and tracking targets step by step. Finally, to improve the trajectory integrity and tracking accuracy, a joint tracking method based on a low-frame-rate tracking criterion was proposed to control the state of the target tracking module. The method was validated on a self-assembled dataset from the GaoFen-4 satellite. The experiment results show the stronger competition and accuracy of the proposed method than other state-of-the-art object trackers.

Index Terms—Deep learning, multilevel supervision network, remote-sensing image sequences, small object tracking.

I. INTRODUCTION

SHIP object tracking from low-resolution remote-sensing images is a challenging task that involves detecting ships, assigning unique ID for each ship target, and tracking their movement across a sequence of satellite images. Although related research studies are not widely studied as a natural visual image target tracking methods, they have demonstrated strong applicability for many fields [1], [2]. Most prevalent ship detection or tracking methodologies currently rely on high-resolution images obtained from satellites in low-Earth

orbit. This approach aims to attain enhanced accuracy in detection or tracking outcomes by extracting intricate details of target shapes and textures [3]. However, low-orbit satellite images have limited coverage and longer revisit cycles, which makes them unsuitable for time-urgent engineering applications. By contrast, high-orbit satellites can acquire wide-swath remote-sensing images with high temporal resolution, which can meet time-urgent engineering applications. Therefore, it has indispensable research value for the development of various fields of society.

Unlike natural images and high-frame-rate remote-sensing image sequences, object tracking on low-resolution remote-sensing image sequences remains a difficult task due to many challenges: 1) low spatial resolution and insufficient target features, leading to increased chances of missing targets; 2) large frame difference time between consecutive frames, resulting in nonoverlapping association of targets and higher possibility of association errors due to the long distance between targets; and 3) remote-sensing images are more susceptible to weather interference such as cloud and fog occlusion, causing target missing and trajectory fracture. These challenges are clearly illustrated in Fig. 1, where Fig. 1(a)–(c) shows the high-frame-rate images and Fig. 1(d)–(f) represents the raw low-resolution and low-frame-rate remote-sensing images. It is evident that this kind of remote-sensing image sequence is very different from high-frame-rate image sequences. Consequently, improving the performance of the model in solving these problems is critical if we want to obtain better tracking results.

At present, object-tracking methods have achieved rapid development with the progress of deep-learning technology. However, in the field of remote-sensing images, target detection and tracking methods have primarily been proposed for high-spatial-resolution images with clear targets. In [4], an optical flow-based tracker was presented to track the vehicles from high satellite video. In [5], a novel feature-attention framework was proposed for high-resolution remote-sensing imagery object detection. Sun et al. [6] proposed a shape robust anchor-free network (SRAF-Net) to detect the garbage dumps in remote-sensing imagery. These methods have demonstrated excellent tracking performance. However, these satellite videos are acquired from low-orbit satellites and possess high spatial resolution. Therefore, these networks are suitable for clear target detection and tracking, but they cannot

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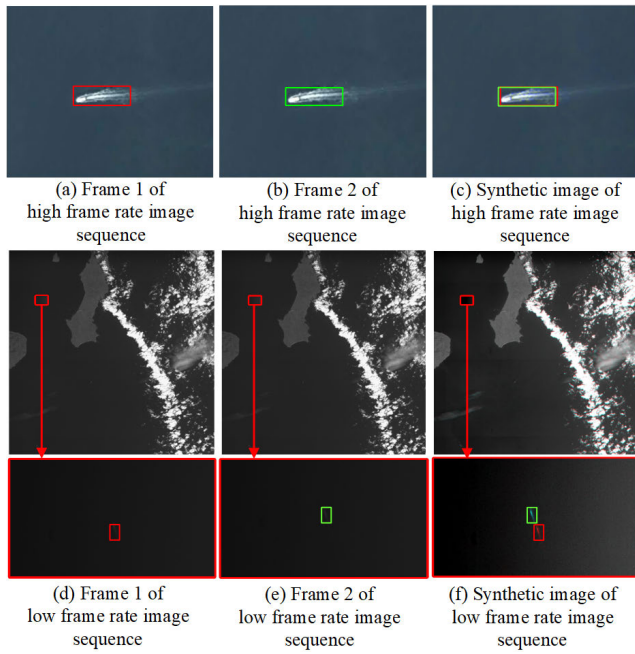


Fig. 1. Comparison of (a)–(c) high-frame-rate image sequence and (d)–(f) low-frame-rate image sequence. The first two columns demonstrate two consecutive frames. The last column of images is the consecutive frame synthetic images of high-frame-rate image sequence and low-frame-rate image sequence, respectively.

achieve the best detection performance for small targets in low-resolution remote-sensing image sequences.

Furthermore, there are also some approaches designed for small object tracking based on natural videos, which can provide some fresh ideas in the remote-sensing community [7], [8], [9], [10]. However, it is important to note that consecutive natural images typically have a short frame difference time, leading to a significant overlap rate among targets. As a result, these methods tend to track objects by capturing similar features between them. Additionally, all of these methods aim to achieve object tracking through a one-step framework with single-level supervision, which can be seen as the blue area in Fig. 2. They improved the model's performance by optimizing the internal network structure. Nonetheless, it is difficult to obtain the optimal solution to a complex problem by a single method. A better idea to solve it is to split a complex problem into multiple subproblems and solve them gradually [11]. For example, Zhang et al. [12] proposed a multistage framework with image dehazing and object detection subnetworks. They used different networks with their own supervision to complete the task level by level, which can achieve better results. However, during training, there is no information interaction and subnetworks are separate from each other. As a result, each network takes the human visual definition as the learning objective and reduces the machine perception effort, which hinders further performance improvement. Consequently, we believe that designing a multilevel supervision network with high integration is crucial for achieving more accurate small object tracking.

In view of the small and dim characteristics of targets in low-resolution remote-sensing images and the above analysis of deep-learning technology, it is crucial to improve the object

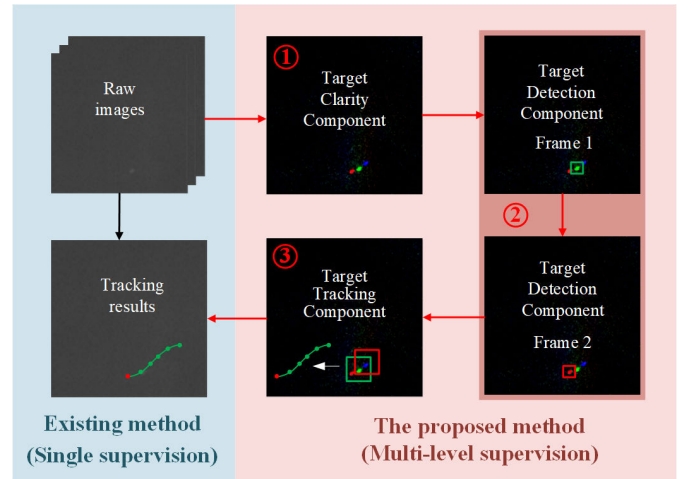


Fig. 2. Pipelines of the existing method and the proposed method. The blue area denotes the most frameworks with a single supervision network and the pink area shows the proposed multilevel supervision network. Red circles indicate the components with supervision.

saliency against the background before inputting the images to the target detection and tracking network. For example, Ma et al. [13] found that the output performance of the network is closely related to the quality of the input image. Additionally, many researchers have made many efforts in the field of image preprocessing, including image enhancement [14], [15], [16], image super-resolution [17], [18], image super-pixels [19], and others. Moreover, there are also some methods to enhance object features by fusing multiple frames [20], [21], [22] and optical flow [23], [24], [25]. By enhancing the images, the insufficient object information can be effectively supplemented. These methods have also shown good performance on remote-sensing images, laying a good technical foundation for the research of multilevel supervision networks. Moreover, our previous work COCO-Net [26] has demonstrated the effectiveness of pre-enhancing target features for object detection.

Building upon previous research, we constructed a novel heterogeneous multilevel supervised network (MULS-Net) for ship tracking based on the tracking-by-detection approach, which integrates multiple network components into an overall model. The main motivation of this method is to improve the dim and small object tracking performance by solving a difficult task step by step. First, we designed a MULS-Net framework to solve the small object-tracking complex task step by step, including the target clarity component (TCC), target detection component (TDC), and target tracking component (TTC). Second, a gradient different-guided target clarification component was utilized to adaptively improve the clarity of ship targets. This component is also implemented based on the multiframe correlation images constructed for supplementing target features. Third, an optimized TTC based on a low-frame-rate tracking criterion was introduced to associate the detection objects to obtain more complete tracking sequences. Besides, a loss-based integration control strategy was designed to enhance the integration performance of different components, making them all learn to improve the final tracking accuracy.

In contrast to the aforementioned research approach [26], although both methods are built upon the same learning paradigm, this research method still has its own uniqueness for small target-tracking tasks. First, this article primarily focuses on the research of tracking small and weak objects. It incorporates a higher-level TTC and modifies the integrated loss feedback strategy. This compositional framework combines both deep-learning and feature-driven methods, further validating the effectiveness and scalability of this multilevel supervised learning paradigm. Meanwhile, it introduces a gradient difference feature extraction module for optimizing the target clarification component specifically for small and weak objects. This enhancement results in a more pronounced improvement in the enhancement of these small and weak targets. Additionally, for the task of object tracking in low-resolution remote-sensing images, a low-frame-rate tracking criterion is proposed to mitigate the impact of cloud and fog occlusions on object tracking, thereby enhancing tracking accuracy. A self-assembled GaoFen-4 remote-sensing image sequences dataset was used to evaluate the effectiveness of the proposed method. Experimental results show that this approach achieves the best tracking accuracy compared to the state-of-the-art trackers.

The contributions of this study are summarized as follows.

- 1) A novel MULS-Net was presented to improve the small ship tracking accuracy. The proposed tracker enables utilizing different components to decompose the difficulty of a single complex task, compensates each other, and achieve maximum tracking performance.
- 2) To adaptively improve the small target saliency, we designed a gradient different-guided target clarification component, which is a simple, yet effective method to highlight ship targets of interest. At the same time, the multiframe correlation images were constructed as input data to improve the target strength.
- 3) A joint object tracking component based on a new low-frame-rate tracking criterion was proposed, which utilizes both deep-learning and feature-driven methods. It not only improves the accuracy of small object tracking, but also effectively reduces the trajectory breaks caused by cloud and fog occlusion.

The rest of this article is organized as follows. Section II investigates the related works, and Section III describes the details of the proposed approach for low-resolution ship tracking. Experimental results and detailed comparisons are shown in Section IV to verify the superiority of our method. Finally, conclusions are drawn in Section V.

II. RELATED WORKS

A. Image Enhancement or Clarification Methods

At present, the enhancement technology for natural images has been widely developed. Early research studies on traditional methods, such as histogram equalization [27] and gamma transform [28], can realize image enhancement simply and directly. However, methods based on the statistical information of images may lead to excessive enhancement or other restrictive problems. The FDHE [29] algorithm introduced a

fuzzy similarity index and fuzzy contrast factor to describe the neighborhood characteristics of pixels, which can enhance the contrast of the image while maintaining the natural characteristics of the image. In addition, deep-learning technology has also shown good application performance in this field. In 2020, a semisupervised learning method based on the deep recursive band network (DRBN) [30] was proposed for low-light image enhancement methods. It was carefully designed to extract a series of coarse-to-fine band representations and produce enhanced results with good reconstruction details with the help of this two-stage design. In [31], an accurate and efficient one-stage object detector based on feature aggregation and augmentation was proposed. It improved the ability to capture features by adding a feature aggregation module and two feature enhancement modules to the SSD network. Although these methods show good enhancement effects on natural images, they are not suitable for remote-sensing images due to the feature differences.

Therefore, some scholars have researched remote-sensing image enhancement processing. Inspired by the earlier dual-domain filter, a progressive dual-domain filter was proposed in [32] to simultaneously enhance and denoise optical remote-sensing images. Experimental results show that the proposed method has good performance in contrast enhancement and noise suppression. To improve the visual intensity of weak and small vehicle targets in remote-sensing images, Gao et al. [33] proposed a goal-guided adversarial generative network algorithm, named DE-CycleGAN. The joint learning and mutual assistance of the gradient enhancement model of the target guidance branch can greatly enhance the target saliency, which provides a good data foundation for subsequent detection tasks. In addition, image clarification methods, such as PSGAN [34] and UPanGAN [35], fused low-resolution multispectral images and high-resolution panchromatic images through optimized generative adversarial networks to achieve the purpose of image enhancement processing. However, due to the problem of insufficient image source data and weak target strength, the application of the above methods in remote-sensing image sequences with weak targets needs to be further optimized.

B. Object-Tracking Methods

In recent years, the target-tracking technology for high-resolution video satellite images has developed rapidly and has been widely used in many fields. In the early stage, generative methods were mainly used to achieve object tracking, such as color histogram [36], HOG [37], SIFT description [38], and HU matrix [39]. Although they are easy and effective for some datasets, while the generative method does not consider more background information, it cannot deal with complex background images well. Therefore, discriminative object-tracking methods have been studied and developed. It works by treating the tracking method as a multiframe detection method. In 2021, a new deep-learning framework [40] combining cross-frame keypoint detection network (CKDNet) and spatial motion information-guided tracking network (SMTNet) was proposed. CKDNet used a custom cross-frame module to

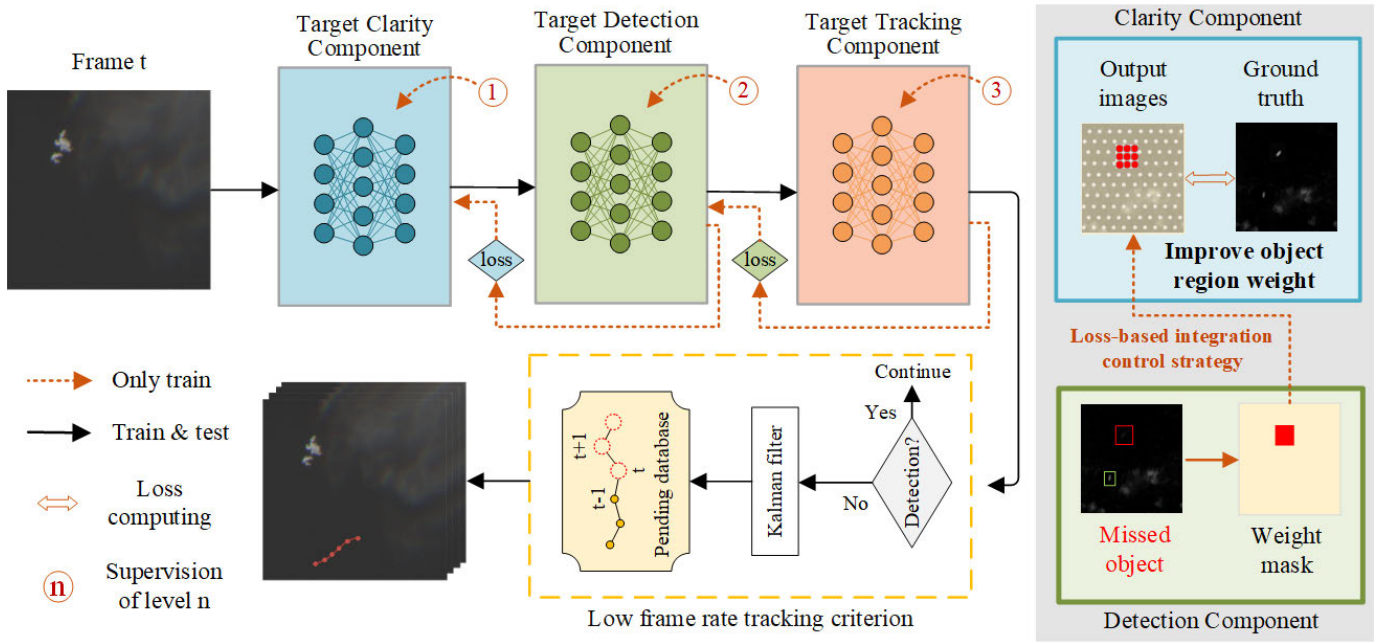


Fig. 3. Whole framework of the proposed multisupervised method Muls-Net. The method consists of three main components: TCC for target clarification, TDC for target detection, and TTC for target tracking.

realize key point detection assisted by complementary information between frames. SMTNet constructed a dual-branch long short-term memory network to achieve accurate dense vehicle tracking on the basis of high-quality detection. In [41], a lightweight parallel network was proposed for object tracking. It not only completed the temporal modeling of video sequences, but also detected moving targets in pixel units, giving full play to the detection and tracking ability of the model for dim and small objects. However, these methods are mainly applied to video satellite images, which still face challenges for low-resolution remote-sensing image sequences with large frame difference times.

On the other hand, high-orbit satellites have distinctive attributes such as large image coverage and short revisit time, which can meet the high timeliness requirements of many engineering applications. In [42], a complete processing algorithm including ship detection and tracking was proposed for low-resolution ocean surveillance satellite images. Potential targets were first detected by constant false alarm rate (CFAR) processing and morphological filtering. In the tracking process, the multiple hypothesis tracking method was used to remove false targets, and finally, a more accurate target tracking result was obtained. Xiao et al. [43] used the frame difference information of image sequence to realize small target detection and tracking. The cloud interference was removed by correlation filtering, and the visual background extraction algorithm was used for target detection. At the same time, the highly reliable discriminant filter was used to achieve effective target tracking. Besides, a ship target-tracking method based on spatiotemporal data fusion and the convolutional neural network [44] was proposed for high-orbit satellite images. The effective spatiotemporal features were used to distinguish the real target from the background interference,

and the appearance consistency measurement method based on structural similarity was used to reduce false alarms. The above methods improve the tracking effect of dim and small targets to a certain extent, but they do not consider the influence of low target contrast in the original image on feature extraction. Therefore, the moving ship tracking method for low-resolution remote-sensing image sequences is still an urgent problem to be solved.

C. Multilevel Supervision Methods

According to experience and experimental knowledge, it is difficult to directly obtain the optimal solution to complex problems through a single model. A better idea is to transform the problem into multiple subproblems and solve them one by one [13]. However, the current deep network models based on multilevel supervision methods have not yet been rapidly developed, and most methods are cascade multimodule networks and independent-supervision combined networks, which have similarities and essential differences with the ideal multilevel supervision network model.

1) *Cascade Multimodule Network*: It is mainly composed of multiple modules for different task purposes, but only through the unique supervised ground-truth (GT) regulation of the overall network to achieve parameter updates. Its main feature is that information transfer and gradient feedback are mainly carried out between modules through feature maps, and the network integration is higher. For example, in [45], a new cascade structure was proposed, including an improved region proposal network (RPN) and an object recognition network. The proposed method can effectively improve the quality of region-of-interest (ROI) proposals without additional computational overhead. At the same time, for the object recognition

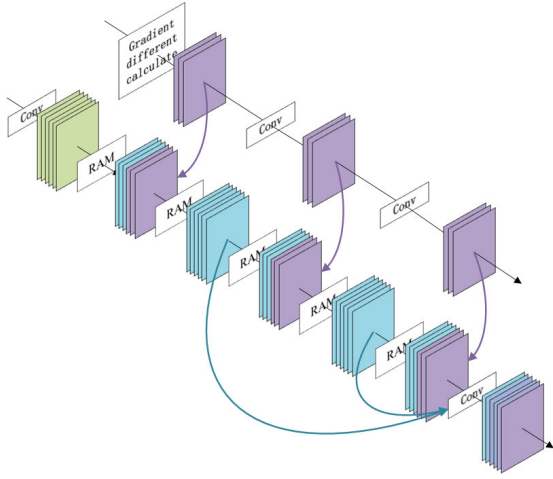


Fig. 4. Schematic of target clarification component. RAMs denote the residual attention modules. Rectangles with different colors represent the feature maps computed by different layers.

module, the global context modeling was optimized, and the performance was greatly improved. The C-RPN algorithm [46], a multistage tracking framework, combined a Siamese network and a cascaded RPN to make full use of multilayer side features and further improve network discriminability. These methods have strong feature coupling ability, but they cannot reduce the difficulty of solving complex problems by the divide-and-conquer strategy, so further network decomposition is needed to achieve the best performance.

2) *Independent-Supervision Combined Network*: Different from the multimodule combination network, this network framework often contains multiple different subnetworks that are supervised by their own. On the one hand, each subnetwork is oriented to different tasks, and there is no information transfer between them during training. For example, the UDnD algorithm [12] proposed a new defogging detection framework. It contains three parts: a residue-aware haze density classifier, a density-aware dehazing network, and a density-aware object detector. Experimental results showed that the multi-stage supervised network can provide high-accuracy detection results. On the other hand, different network components are oriented to different scales of the same task. Kansal and Nathan [47] proposed a novel super-resolution architecture, which generated super-resolution images at $\times 2$, $\times 3$, and $\times 4$ scales by introducing supervision at different feature layers, so that a single model can deal with tasks at multiple scales. Experimental results evaluated that the model achieves state-of-the-art results on the PBVS 2020 dataset. Although these methods have improved the model effect to a certain extent, the intensity of machine perception has been reduced in the network because the learning goal of each component is defined artificially, which limits further performance improvement.

III. METHODOLOGY

A lot of object-tracking methods have enhanced target features by fusing the multiscale features or utilizing multi-frame motion information to compensate for object features. Nonetheless, a key point is that the relationship between network input and model performance is tight, which is

important but easily overlooked in the field of remote-sensing image processing. Therefore, a multilevel supervised ship tracking model was constructed in this study, which was inspired by a new concept that a difficult task requires decomposing operations to reduce the difficulty. Specifically, the whole framework consists of three components with their own supervision, the target clarification component, the TDC, and the TTC, which can be seen in Fig. 3. It is described in detail below.

A. Target Clarification Component

Because the ship object in the single-band remote-sensing image is dim and small, we used three consecutive frames of images to compensate for and enhance the target features. Assume that the consecutive frames are $I_t - 2$, $I_t - 1$, and I_t , they were synthesized as a three-channel image by stacking and input the TCC network, which is illustrated as Fig. 4. It mainly consists of five residual attention modules (RAMs) and introduces gradient difference features for optimization. Each RAM contains a channel attention block and a spatial attention block, which are internally connected by residual skip connections and inspired by the method in [48]. The output features of several RAMs were fused by a convolution layer at the end, which can be expressed as

$$\delta_e = c \left(\prod_{i=1}^n R(I_i \oplus I_i^g) \right) \quad (1)$$

where δ_e denotes the calculation of the TCC. c represents the convolution layer and $\prod$ represents the cascade of multiple layers of computational modules. R represents the RAM module. I_i and I_i^g represent the i th input feature and the gradient difference map, respectively. $\oplus$ denotes the concatenation of different features.

Among them, the novel and interesting gradient difference operator is developed, which can be defined as

$$I_x^g = I_i - I_{i-1}, \quad i = 1, \dots, h \quad (2)$$

$$I_y^g = I_j - I_{j-1}, \quad j = 1, \dots, w \quad (3)$$

$$I^g = c(I_x^g \oplus I_y^g) \quad (4)$$

where I_x^g , I_y^g , and I^g are the gradient difference maps on the x - and y -directions and the obtained feature map, respectively. h and w denote the height and width of the input image, respectively. The gradient difference feature maps have a clearer feature of dim and small targets, as shown in Fig. 5 and further enhance the learning ability of the network and reduce the missed detection rate of the model.

B. Target Detection and Tracking

1) *Target Detection*: Relying on the obtained clear images with salient objects, the detector optimized from the one-stage method is utilized to detect the ship objects. This ship detector consists of a backbone, a neck, and a head, which are key modules in most one-stage detection algorithms. There are three blocks in the backbone to extract object features, each of which consists of five residual convolution units. The neck module is used to fuse the features with different scales,

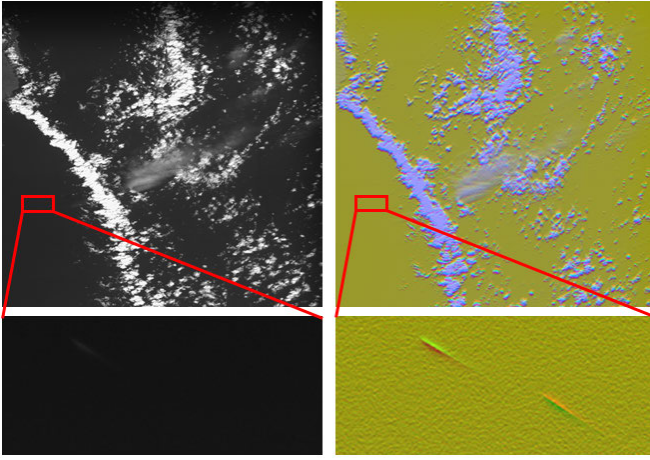


Fig. 5. Schematic of the (left) raw remote-sensing image and the (right) gradient different map.

and the network framework design also follows the structure of a feature pyramid network (FPN) and a path aggregation network (PAN). FPN uses top-down side connections to construct high-level semantic features at all scales, and PAN incorporates bottom-up routes to compensate for and strengthen the localization information. They can be learned in [49]. The algorithm calculation formula is shown as follows:

$$\delta_d = \phi \left(\varphi \left(\prod_{i=1}^3 (b_i(I)) \right) \right) \quad (5)$$

where ϕ and φ represent the operators of the detection head and neck, respectively. b_i is the i th feature extraction block in the backbone and I is the input image.

2) *Target Tracking*: It is achieved in TTC by associating the detection results of different frames, including the deep-learning method and feature-driven method. On the one hand, a convolution neural network was designed to extract the object features, including the detected targets at the current moment and the tracked targets from the previous moment. This network is composed of eight basic convolution blocks, including four downsampling residual convolution blocks and six nondownsampling residual convolution blocks. Notably, since the input images are composed of three consecutive frames, there inevitably are two same frames in the images fed into the network continuously. For example, both image1 (I_{t-2}, I_{t-1}, I_t) and image2 (I_{t-1}, I_t, I_{t+1}) contain frames I_{t-1} and I_t . Therefore, it is a good tracking strategy to utilize the feature of the same frame in the input images to achieve the correlation of consecutive targets. Based on this, the first two frames of the input image at the current moment and the last two frames of the input image from the previous moment were used as the basic images for the target feature extraction network. In addition, we used the expanded target region as the input of the feature extraction network, which has a side length ratio of 3:1 to the original detected target region. After obtaining the target feature vectors through the network, the cosine distance between them is calculated to determine the target similarity and the best tracking result is obtained through the Hungarian matching algorithm.

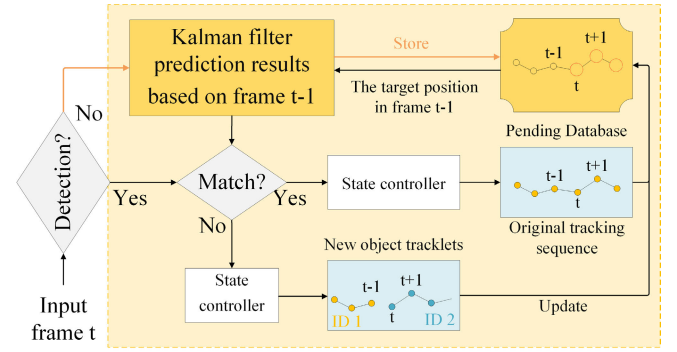


Fig. 6. Schematic of the developed low-frame-rate tracking criteria. The yellow line represents the flow of the tracker when the current frame does not detect the target.

On the other hand, the feature-driven method is used to handle detections and tracks that are not matched through the above deep-learning method. For these tracks, use Kalman filters to predict each target's current position according to the target position in the previous moment. Next, the square Mahalanobis distances between the current targets and the predicted targets are calculated to generate a cost matrix. The Hungarian matching algorithm is then applied to obtain the best-matched results and determine whether these targets can be successfully matched the second time. Finally, the status of the target tracker is updated based on the matching results. This feature-driven target-tracking method is a commonly used algorithm at present, which is not repeated in this article, and the specific content can be found in [50].

C. Low-Frame-Rate Tracking Criterion

Due to the cloud cover phenomenon, the ship object may not be detected during motion in a certain frame, which can cause the tracking trajectory to break. To solve this problem, we proposed a low-frame-rate tracking criterion to reduce the impact of cloud cover on target tracking as much as possible. Specifically, we presented temporarily storing the Kalman prediction result in the "Pending database" when a target is not detected in a frame, as shown by the yellow line in Fig. 6. This stored result can be used as a tentative target detection result for the current frame. When this target is detected again in the subsequent frame, it is matched with the Kalman prediction result generated from the previous frame. At this time, it needs to be judged in two cases: 1) if the target can be matched, the new target ID is still defined as the original target ID, and the Kalman filter results of this target in each frame during the occlusion are taken as the determined object boxes, and the trajectory fusion is completed and 2) if the target cannot be matched, it is assigned a new target ID and distinguished from the original target sequence. At the same time, the relevant results stored in the database are deleted. Generally, the number of frames in which the target is occluded by small clouds during motion is less than 5 in low-resolution remote-sensing images excluding cases of long-term occlusion. Therefore, if a target is lost for more than five consecutive frames, the sequence ID is also deleted in the pending database and no longer matched with the newly

detected target. The detail of the trajectory criteria can be found in Fig. 6.

D. Loss-Based Integration Control Strategy

In the MULS-Net, each component has its own GT and loss function to regulate the network. First, the mean absolute error is used as the loss function for TCC, which aims to obtain the difference for all the pixels in the image pair

$$L_e = \frac{1}{m} \sum_{i=1}^m |G_i - P_i| \quad (6)$$

where G and P stand for GT and predict the result. i represents the i th pixel and m is the sum number of the pixels. For TDC, the popular focal loss is utilized as a classification loss function. Then, the total loss is obtained by weighted summation of the classification loss (L_1), intersection over union loss (L_2), and confidence loss (L_3)

$$L_d = \lambda_1 \cdot L_1 + \lambda_2 \cdot L_2 + \lambda_3 \cdot L_3. \quad (7)$$

In addition, the correlation loss strategy is used to link all the components, to make them all improve the attention for ship targets. First, the tracking results information of TTC is fed back to TDC for adjusting its loss function. Specifically, the tracking results information is represented by unmatched detections, unmatched tracks, and matched objects with incorrect IDs. It can be expressed as follows:

$$L_t = \frac{UD + UT + \phi}{GT} \quad (8)$$

where UD and UT represent the number of unmatched detections and the unmatched tracks. ϕ denotes the number of objects that are matched but with incorrect IDs. GT is the number of all truth objects. Then, the final loss function of TDC is modified as

$$L_d = \lambda_1 \cdot L_1 + \lambda_2 \cdot L_2 + \lambda_3 \cdot L_3 + \lambda_4 \cdot L_t. \quad (9)$$

With this control strategy, the tracking information is fed back to the detection network, enabling it to improve not only its detection ability, but also its ability to serve the final tracking task. A large number of experiments and studies have shown that the network works best when the weights of these loss function terms are balanced [49]. Therefore, all parameters λ_n are taken as 1 in this study.

Furthermore, the target boxes obtained from TDC are treated as the target mask and feedback to the TCC, which is also utilized to adjust the loss function and makes the network focus more on the target area when learning. As a consequence, L_e are modified to

$$L_e = w_1 \sum_{\Omega} |G_i - P_i| + w_2 \sum_{1-\Omega} |G_i - P_i| \quad (10)$$

where Ω stands for the target mask region. w_1 and w_2 denote the weight factors, respectively. Based on previous experimental experience, when w_1 and w_2 are set to 0.8 and 0.2, respectively, better results are achieved [26]. Therefore, these two parameters are set to 0.8 and 0.2.

IV. EXPERIMENTS

In this section, we verified the effectiveness of the proposed method through experiments. First, we described the experimental conditions, including the dataset used for experiments, the specific implementation details, and evaluation metrics. Second, the results of ablation experiments for the proposed method were discussed. Finally, the comparison experiments with SOTA methods were conducted and analyzed.

A. Experimental Conditions

1) *Dataset*: We conducted experiments based on sequences of low-resolution remote-sensing images from the GaoFen-4 satellite as shown in Fig. 7. It can capture continuous images of the same area with a frame interval of 20 s. The experimental images are single-band visible-light images with a spectral range of 0.45–0.52 μm and a spatial resolution of 50 m. The dataset used for training and testing contains image patch sequences of size 256×256 . It has 645 training sequences, 162 validation sequences, and 200 test sequences. Furthermore, due to the low image resolution, our targets are mainly various types of aircraft carriers and large warships. Their size range is about 120–300 m, which is about 2–6 pixels in a single-frame image, and falls into the category of dim small targets. Therefore, it is a unique dataset for small object tracking in remote-sensing sequences. Moreover, the GT of the ship targets is labeled by professional ship target interpreters, which has high credibility.

2) *Implementation Details*: We implemented the proposed approach based on the PyTorch framework and trained it on an NVIDIA GeForce RTX 3090 24G GPU workstation. The whole network was trained for 500 epochs until the loss remained stable at the end. To obtain the best performance for the detection of small-size objects, we set the IoU threshold to 0.2, the confidence threshold to 0.3, and the NMS threshold to 0.5. The model was optimized by the Adaptive Moment Estimation (Adam) optimizer. The initial learning rates for the object augmentation component and object detection component were set to 0.00001 and 0.001, respectively. Both of them used the cosine annealing strategy to adjust the learning rate.

3) *Evaluation Metrics*: The tracking performance was evaluated by two classes of metrics. One is the common metric, multiple-object tracking accuracy (MOTA) [51], which can be seen as follows:

$$\text{MOTA} = 1 - \frac{\text{FN} + \text{FP} + \text{IDsw}}{\text{GT}} \quad (11)$$

where FN and FP represent the number of false negatives and false positives. IDsw represents the number of identity switch errors, where the tracking algorithm assigns incorrect identities to objects. GT represents the total number of GT objects. MOTA offers a simple and widely used overall measure of tracking accuracy. The other is the new popular metric, higher-order tracking accuracy (HOTA)

$$\text{HOTA} = \sqrt{\frac{\sum_{c \in \{\text{TP}\}} A(c)}{\text{TP} + \text{FN} + \text{FP}}} \quad (12)$$

$$A(c) = \frac{\text{TPA}(c)}{\text{TPA}(c) + \text{FPA}(c) + \text{FNA}(c)} \quad (13)$$

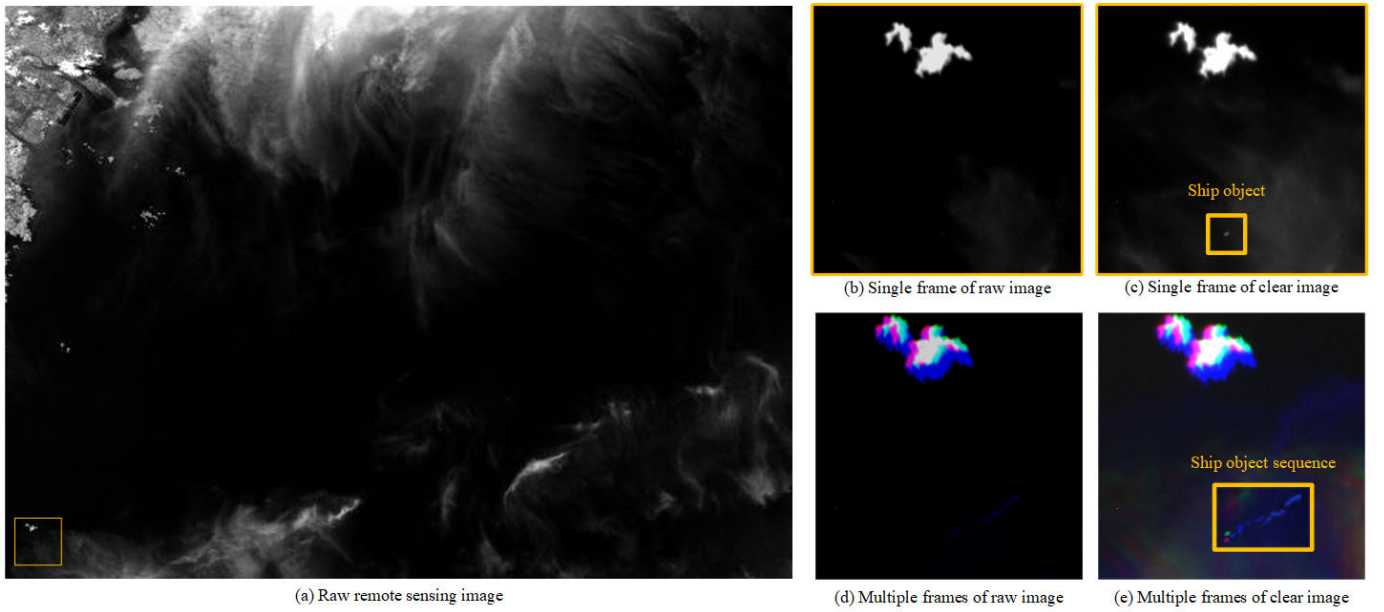


Fig. 7. Example images of raw low-resolution satellite dataset. (a) Raw remote-sensing image. (b) and (c) Single-frame images. (d) and (e) Image integrating consecutive multiframes. (c) and (e) Images with enhanced targets. Meanwhile, the bright spots in (c) and (e) are ship targets. Because (e) is displayed by a three-band RGB image, the ship target sequence shows red dots, green dots, and blue dots, respectively.

where $TPA(c)$ represents the number of true-positive associations, indicating correctly associated object tracks. $FPA(c)$ and $FNA(c)$ represent false-positive (FP) associations and false-negative (FN) associations, respectively. For more detailed information, refer to [52]. HOTA provides a more detailed and comprehensive evaluation of multiobject tracking performance, considering both detection and association aspects.

In addition, IDF1 Score (IDF1) and Association (AssA) are used to measure association performance, and disentangling the performances for detection (DetA) is utilized for detection evaluation

$$IDF1 = \frac{2TP}{2TP + FP + FN}. \quad (14)$$

It is a combination of precision and recall

$$AssA = \frac{1}{TP} \sum_{c \in \{TP\}} A(c) \quad (15)$$

$$DetA = \frac{TP}{TP + FP + FN}. \quad (16)$$

Both AssA and DetA contribute to the overall evaluation of multiobject tracking algorithms, with higher values indicating better tracking and detection accuracy. For ease of analysis, widely used statistical results are also provided, such as the number of FNs, FPs, and the percentage of mostly tracked (MT) trajectories and mostly lost (ML) trajectories.

B. Analysis of Different Scheme Settings

Here, we ablate the core design scheme of the proposed method. For fair validation, the parameters of each model were kept as consistent as possible. The models were trained and validated on different parts of the self-assembled GaoFen-4 dataset.

TABLE I

COMPARISON ON THE METHODS WITH DIFFERENT SCHEME.
* REPRESENTS THE COMPONENT WITHOUT THE OPTIMIZED
MODULES AND $\uparrow$ INDICATES THE OPTIMIZATION
DIRECTION OF THE INDICATOR

Scheme	HOTA $\uparrow$	MOTA $\uparrow$	AssA $\uparrow$	DetA $\uparrow$
Baseline	61.4	75.7	60.5	63.3
TDC + TTC*	64.3	78.2	62.7	66.6
TCC* + TDC + TTC*	67.9	81.1	66.9	71.2
TCC + TDC + TTC*	70.3	83.5	68.5	72.8
TCC + TDC + TTC	72.4	87.8	69.6	74.5

1) *Analysis of the Multilevel Supervised Framework:*
Table I gives the performance comparison of each module or scheme proposed in this article. The baseline is the method with Yolov5 as the detector and the traditional Kalman tracking method as the tracker. Row #2 stands for the framework constructed by the proposed TDC and TTC* without a low-frame-rate tracking criterion (TTC*). Through the comparison of these two rows, it is obvious that the proposed tracking method is of great help in improving the tracking performance of the small object. The third row adds the target clarification component (TCC*) before detection, while this component is the model without the gradient different branch. It produces better detection and tracking performance than the previous model with +3.5 HOTA and +2.9 MOTA, respectively. In addition, the comparison (row #3 versus row #4) shows that the addition of the gradient difference feature can produce a positive effect on the tracking performance (+2.4 HOTA and +2.4 MOTA). The last row denotes the method that adds the loss-based integration control strategy to the former, which also represents the final proposed complete model

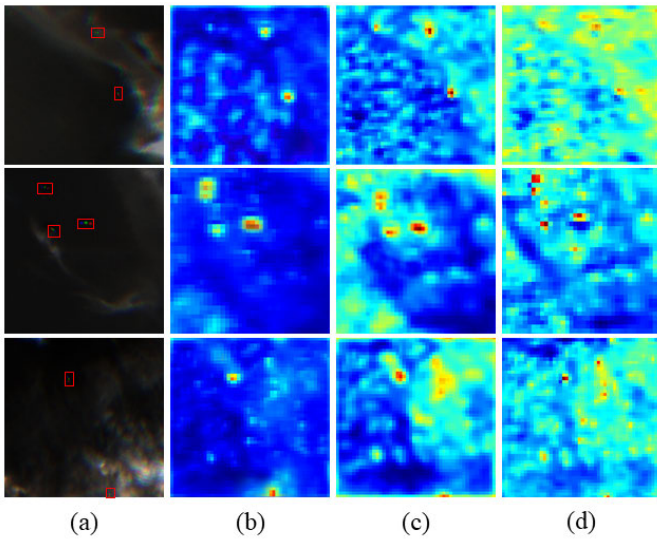


Fig. 8. Hotmaps of different models. (a) Enhanced images. (b)–(d) Feature maps of the MULS-Net, MULS-Net without gradient different guidance, and MULS-Net without TCC, respectively.

(MULS-Net). The best result on the GaoFen-4 dataset can reach 72.4 HOTA and 87.8 MOTA, which is also sufficient to show that all proposed designs are valid for small ship tracking on low-resolution remote-sensing sequences.

In addition, the feature maps visualized in the detectors of different models are shown in Fig. 8. To be able to observe more clearly, the pseudo-color images are illustrated in Fig. 8(a), constructed based on three consecutive frames. At the same time, we enhance the target areas, because the targets are difficult to be seen in the raw images. The images in Fig. 8(b)–(d) are the hot maps of the MULS-Net (corresponding the last row of Table I), MULS-Net without gradient different guidance, and the MULS-Net without TCC, respectively. Compared with other models, the targets of the feature maps in MULS-Net are more prominent and the background is relatively smooth and clean. The images in the fourth column contain more nontarget noises, which may be caused by insufficient target information due to the lack of TCC in the network. These samples indicate more intuitively that the proposed modules are indeed useful.

2) *Analysis of Gradient Different Optimization:* This section focuses on finding the optimal parameters for the number of gradient different features (α) introduced in TCC. Fig. 9 explains the corresponding experimental results. It can be found that the tracking indices improve with the increase in the number of gradient difference features introduced. However, when α is less than 3, the performance improves faster. When α is greater than 3, the performance increase flattens out. Therefore, to balance the parameter volume and tracking performance of the network model, the number of gradient difference features in this study was set to 3.

C. Detection Results and Comparison

In this section, we compared the proposed method, MULS-Net, with other state-of-the-art trackers on the GaoFen-4

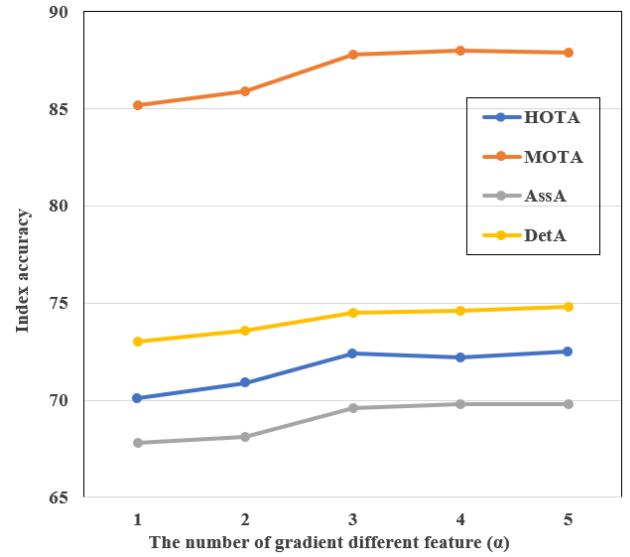


Fig. 9. Visualization results based on the comparative experiments of gradient different map number α .

dataset. These methods include some recent track-by-detection methods, such as DAN [53], Tracktor + CTdet [54], DeepSORT [50], and RAN [56], and some popular joint-detection-and-tracking methods, including TubeTK [55], CTracker [57], CenterTrack [58], and TraDes [59]. Table II shows the quantitative evaluation results, where the optimal results are shown in red bold and the second-best results are shown in blue font. The proposed approach outperforms the TraDes, the best-performing method among the other compared methods, results by 4.5 HOTA and 3.6 MOTA, respectively. The IDF1 reaches 88.5, which is also higher than that of the other trackers investigated. Looking at the experimental results of all methods, the method based on the joint-detection-and-tracking framework is relatively better than the other methods.

It is worth noting that there was a breakthrough in our approach. Compared to the best track-by-detection method RAN, the present approach produces an approximately 11% improvement in HOTA and 9% in MOTA. Moreover, we collect the number of FNs and FPs to prove the effectiveness of each method. It can be seen that for such low-resolution remote-sensing image sequences, our method contains the least number of missed detections and false alarms. The visual target boxes for the proposed method are illustrated in Fig. 10. The pseudo-color image synthesized from long consecutive frames and the binarized image are also given, which can make the target trajectory more clear. The proposed network model accurately tracks each target in these test images with fewer missing targets and target ID changes, as can be observed in the results.

Furthermore, Fig. 11 illustrates the tracking results of the compared methods. To achieve better result comparison, we selected large-scale images (1200×1200 pixels) containing multiple targets for testing purposes. The detection boxes of our approach are displayed on the R-G-B images

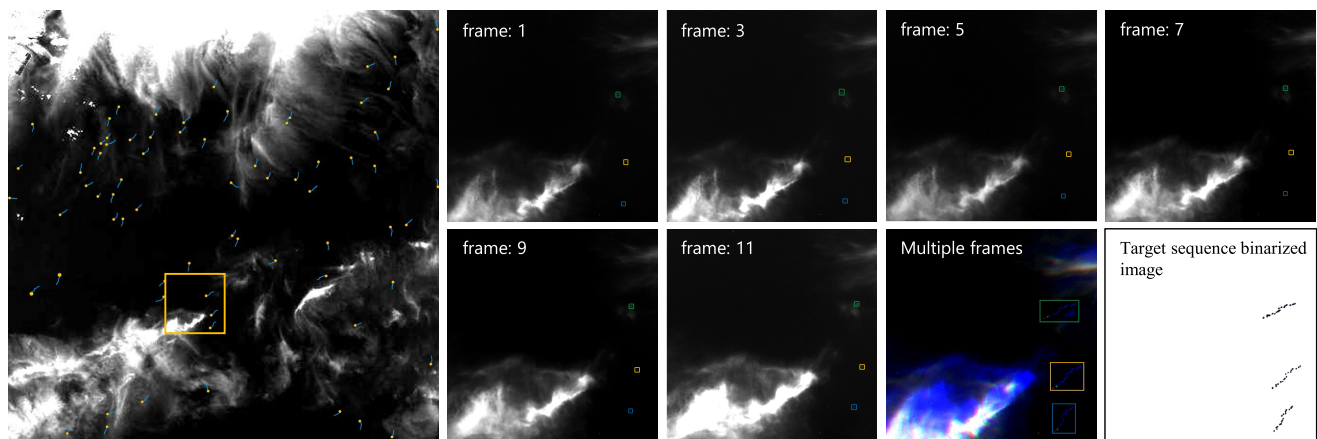


Fig. 10. Tracking results of the proposed method. (Larger image on the left) Multiple object tracklets within 6 min. (Subimages on the right) Detection results of different frames, the pseudo-color image synthesized from these consecutive frames, and the binarized image of target trajectories.

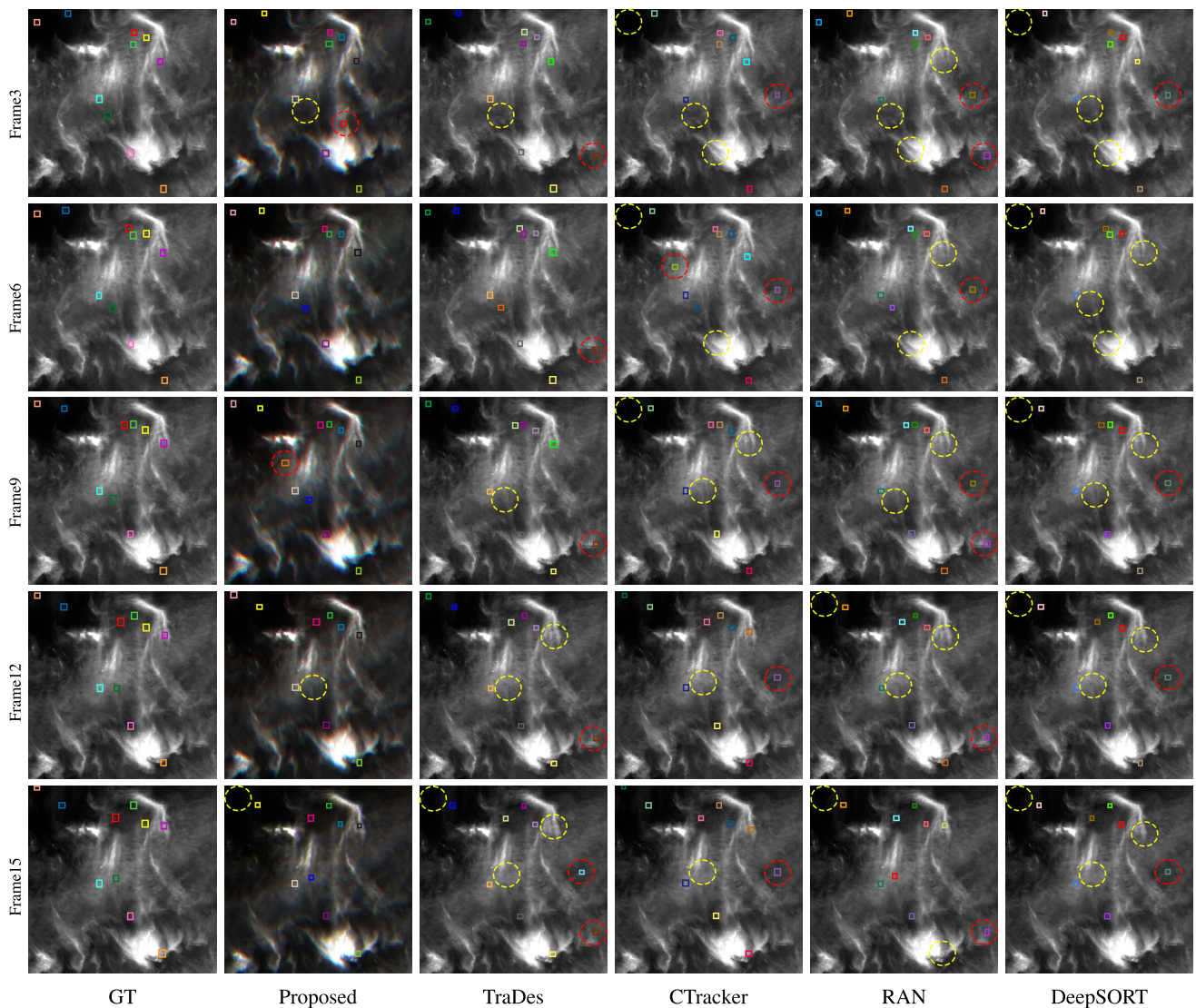


Fig. 11. Tracking results of the compared methods in different frames, where GT denotes the ground truth and the last several columns are compared methods. Boxes with different colors represent different ship targets. The bright spots in the detection boxes represent the ship targets, which contain only a few pixel values in size. The red circles represent FP targets, while the yellow circles represent missed detection targets.

synthesized from three consecutive frames. The results of other methods are displayed on the single-band image after linear

stretching so that the target has higher visual saliency. Due to the low image resolution, the ship objects only consist

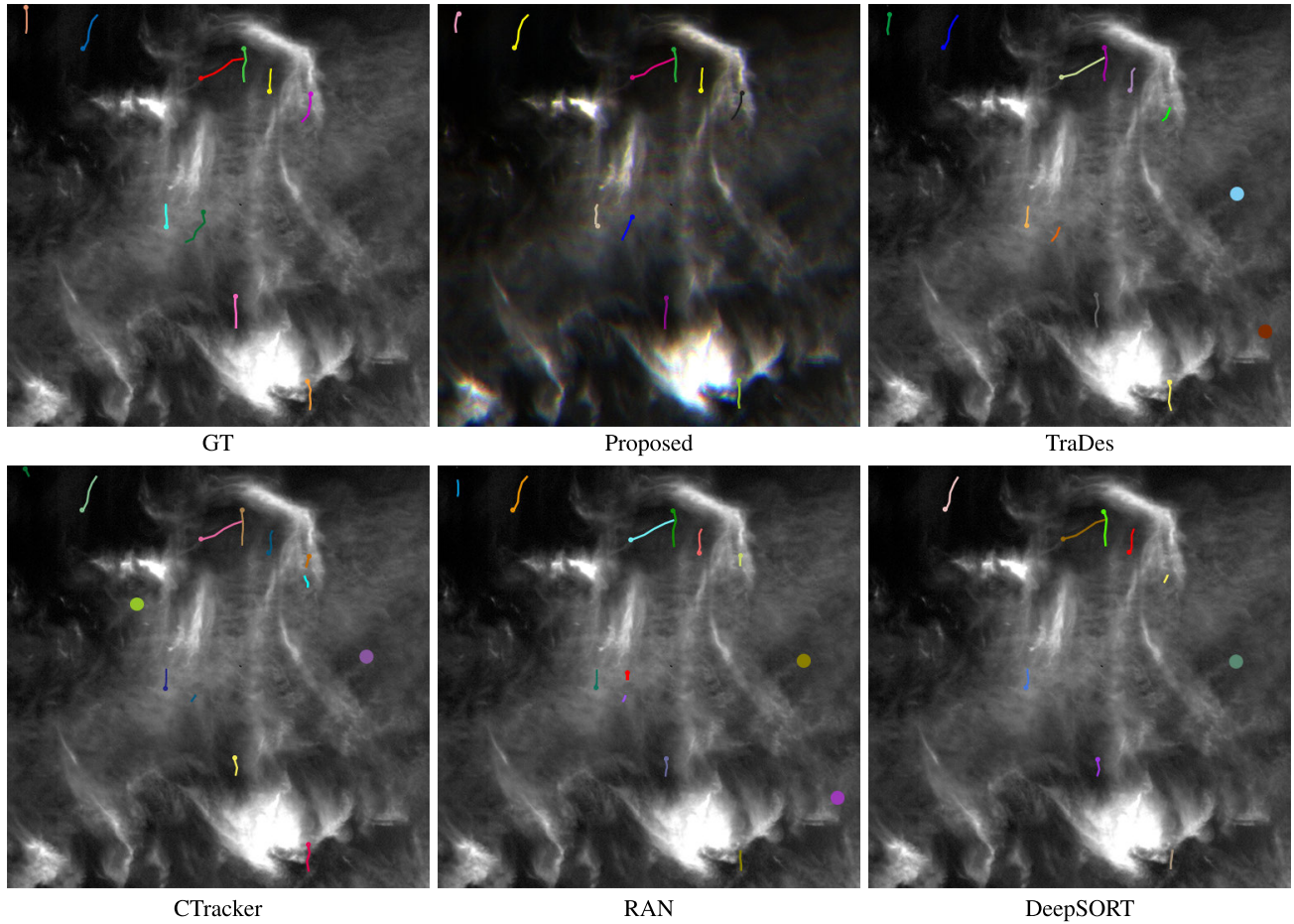


Fig. 12. Tracking trajectory of the compared methods shown based on frame 15. Lines with different colors represent different ship target trajectories. Among them, the larger origin is the false alarm. In some methods (like CTracker and RAN), there are trajectories with breaks and color transition, which indicates that the target ID has changed during the tracking process.

TABLE II
EVALUATION INDICES OF DIFFERENT METHODS BASED ON THE GAOFEN-4 DATASET. BOLD LETTERS REPRESENT THE BEST RESULTS.
↑ AND ↓ INDICATE THE OPTIMIZATION DIRECTION OF THE INDICATOR

Method	HOTA↑	MOTA↑	IDF1↑	AssA↑	DetA↑	MT↑	ML↓	FN↓	FP↓
DAN [53]	59.1	73.2	72.6	57.3	60.2	55.8	17.7	1459	3982
Tracktor+CTdet [54]	59.9	74.0	73.1	58.3	60.9	56.7	18.2	1428	4002
DeepSORT [50]	60.4	75.3	73.9	59.7	61.8	57.3	16.9	1294	3937
TubeTk [55]	60.7	77.1	76.2	58.9	62.4	57.1	14.5	1521	3409
RAN [56]	60.2	75.4	75.8	59.7	61.6	58.4	17.9	1402	3911
CTracker [57]	64.4	81.9	80.8	63.7	67.6	59.7	16.8	1398	3822
CenterTrack [58]	65.8	82.3	83.1	64.4	68.8	61.7	16.2	1223	3789
TraDes [59]	67.9	84.2	85.3	66.3	69.2	63.4	15.9	1432	3321
MULS-Net(Ours)	72.4	87.8	88.5	69.6	74.5	68.3	12.3	1084	2571

of a few pixels. Then, they appear as bright spots in the image, similar to the bright spots within the object boxes in the GT of Fig. 11. We used square boxes to represent tracked targets in each frame, red circles for false alarms, and yellow circles for missed detections. We can see intuitively that the joint-detection-and-tracking methods have relatively

higher tracking performance than the other track-by-detection methods, but this does not mean that they perform best for every task. For small object-tracking tasks in low-resolution remote-sensing images, the proposed method shows a higher detection rate and lower false alarm rate. Additionally, the tracking trajectories of compared methods are illustrated in

Fig. 12. It can be observed that the trajectories generated by the proposed method are more complete. Even if there is a frame in the tracking process where the target is not detected, it does not affect the accuracy of the subsequent tracking. At the same time, it also demonstrates fewer false alarms, which are not retained during the tracking process. Therefore, we can conclude that the proposed multilevel supervised method exhibits stronger robustness against cloud interference and better tracking performance for the task of small and dim targets tracking in low-resolution remote-sensing images.

D. Discussion

The proposed MULS-Net provides a novel learning framework for small object-tracking tasks. It achieves step-by-step solutions to complex tasks by dividing a single complex network into multiple subnetworks. In addition, it also reduces the learning difficulties of the single supervision network by introducing more empirical knowledge. However, the proposed method still faces some problems that require further breakthroughs. On the one hand, as the proposed method mainly focuses on detecting the moving ship targets in the ocean background, its detection effect is not good enough for stationary targets or moving targets with very slow speeds. At the same time, due to the small size of the target, the features of stationary targets are closer to the scattered cloud, which increases the difficulty of discrimination. On the other hand, the introduction of multilevel supervision brings a relatively larger computational cost. Therefore, developing lightweight methods for MULS-Net architectures is crucial and will be the focus of our future research. Nevertheless, the MULS-Net still provides a new and meaningful research perspective for the remote-sensing image-processing field.

V. CONCLUSION

This work presented a multilevel supervised tracking model from a new perspective, focusing on achieving accurate small object tracking on low-resolution remote-sensing image sequences through accomplishing subtasks step by step. This approach comprises three components: TCC, TDC, and TTC. Guided by the simple, yet effective gradient different map, the TCC learns to adaptively improve the saliency of weak and small objects. Then, the TDC takes the clearer images from TCC as inputs and detects the objects in consecutive frames separately. Finally, the TTC based on a low-frame-rate tracking criterion is utilized to match the targets of the consecutive frames, which can reduce the interference of cloud cover and improve the tracking trajectory integrity. Exhaustive experiments and ablations illustrate the superiority of our approach. This hierarchical framework for processing difficult tasks can be extended to multiple application fields. However, it is worth noting that an overall regulation mechanism is necessary in this framework to motivate different components toward the same goal, which is the foundation for the MULS-Net. In the future, we will conduct further research on the lightweight models and aim to propose a fast and accurate tracking model, which is attractive for real applications.

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